Residual-Aware Bayesian Clustering for Local Dynamical Models

A Unified Framework for Piecewise Linearization of Vector Fields

Lorenzo Tomaz
AE Studio

Abstract

We formulate the piecewise linearization of a dynamical system $\dot{x} = f(x)$ as a Cluster-Weighted Model (CWM)—a mixture model over the *joint* density of state and vector-field observation, $p(x, f(x)) = \sum_k \pi_k \, p(x|k) \, p(f|x, k)$. Unlike standard GMM (which clusters by geometry alone) or mixture-of-experts (which models the conditional $p(f|x)$ without clustering inputs), the CWM formulation assigns points to clusters based on both geometric proximity and local-model prediction quality. We extend the CWM framework—previously developed for statistical clustering and classification—to dynamical systems, with three local-model parameterizations: a physics-constrained Taylor variant using the analytic Jacobian (zero free local parameters), a data-driven least-squares affine fit, and a Koopman-theoretic variant where each cluster carries a continuous-time EDMD generator in a lifted polynomial space. We derive the EM updates for all three variants, show they share a common E-step and differ only in the M-step, and present empirical results on two canonical dynamical systems—the Lorenz attractor and the damped pendulum. On Lorenz, global EDMD with a degree-2 polynomial lift is exact, so no clustering method can improve upon it—but residual-aware clustering is a strict, statistically significant improvement over standard GMM ($p < 0.001$). On the pendulum, where the nonlinearity $\sin(\theta)$ has no finite polynomial representation, the framework delivers an order-of-magnitude parameter-efficiency gain over global EDMD ($p < 0.001$). All results are validated across 10 random seeds with paired t-tests and 95% confidence intervals.

1 Introduction

1.1 The Partition Problem

Consider a continuous dynamical system

$$\dot{x} = f(x), \qquad x \in \mathbb{R}^d, \qquad f : \mathbb{R}^d \to \mathbb{R}^d. \tag{1}$$

In many applications—reduced-order modeling, control design, stability analysis, reachability, surrogate modeling—one wishes to approximate f with a simpler function that is tractable to analyze or cheap to evaluate. The simplest such approximation is a single linear model $f(x) \approx A(x - x_0) + b$ built from the Jacobian $A = \partial f / \partial x|_{x_0}$. This works near x_0 but the approximation error grows as $O(\|x - x_0\|^2)$, so a single global linearization is inadequate for any problem whose trajectories explore a large portion of state space.

The natural remedy is **piecewise linearization**: partition the region of interest $V \subset \mathbb{R}^d$ into N subregions $V_1, \ldots, V_N$ with centers $c_k \in V_k$ and local models $L_k(x) = A_k(x - c_k) + b_k$. Two questions arise: *where should the c_k be placed?*, and *how should phase space be partitioned into the V_k?* A naive answer: cluster $\{x_i\}$ geometrically (k-means, GMM) and evaluate the Jacobian at each centroid. But this ignores the dynamics. A region with high sample density but high *curvature* of

f is a bad choice of subregion. *The partition should be chosen not by geometric proximity but by **linearization quality**.* Centers should sit where f is locally flat; boundaries should fall where f changes character.

1.2 A Generative Framing via Cluster-Weighted Models

The natural probabilistic framework for jointly clustering inputs and fitting local regressions is the **Cluster-Weighted Model** (CWM), introduced by Gershenfeld [5] and formalized statistically by Ingrassia et al. [7, 8]. A CWM models the joint density of an input-output pair (x, y) as

$$p(x, y) = \sum_{k=1}^{N} \pi_k \, p(x \mid z = k) \, p(y \mid x, z = k), \tag{2}$$

factoring each component into a *marginal on inputs* and a *conditional regression on outputs*. In the Linear Gaussian CWM [7], both are Gaussian: $p(x \mid z = k) = \mathcal{N}(x; \mu_k, \boldsymbol{\Sigma}_k)$ and $p(y \mid x, z = k) = \mathcal{N}(y; \beta_{k,0} + B_k x, \Gamma_k)$.

We recognize that the piecewise-linearization problem is naturally a CWM with x = state vector and $y = f(x)$ = vector-field observation:

$$p(x_i, f(x_i) \mid z_i = k) \;=\; \underbrace{\mathcal{N}(x_i;\, c_k, \boldsymbol{\Sigma}_k)}_{\text{proximity}} \cdot \underbrace{\mathcal{N}(f(x_i) - L_k(x_i);\, 0,\, \sigma^2 I)}_{\text{residual}}. \tag{3}$$

The proximity factor models where each cluster sits in phase space. The residual factor models how well the local dynamical model L_k predicts $f(x_i)$—a point is assigned to component k only if both factors agree. This joint criterion makes the clustering dynamics-aware: boundaries follow linearizability contours rather than density contours.

While the joint-likelihood structure is inherited from CWM, the specific local-model parameterizations we propose—particularly the physics-constrained Taylor variant and the Koopman-theoretic EDMD variant—are novel and have no precedent in the CWM literature, which has focused on statistical classification and regression rather than dynamical systems.

Writing $\varepsilon_k(x) = f(x) - L_k(x)$:

$$\log p(x_i, f(x_i) \mid z_i = k) = \text{const} - \tfrac{1}{2}(x_i - c_k)^\top \boldsymbol{\Sigma}_k^{-1}(x_i - c_k) - \|\varepsilon_k(x_i)\|^2/(2\sigma^2). \tag{4}$$

The hyperparameter σ^2 sets the scale at which residuals matter: $\sigma^2 \to \infty$ reduces to standard GMM; $\sigma^2 \to 0$ makes the framework a piecewise regression.

1.3 Contributions

1. We identify piecewise linearization of vector fields as a **Cluster-Weighted Model** [5, 7]—connecting the dynamical-systems problem to a well-studied statistical framework—and extend it with domain-specific local-model parameterizations.

2. A **physics-constrained variant** (Taylor-analytic) where the local regression parameters are set by the analytic Jacobian $\boldsymbol{J}(c_k)$, yielding zero free local-model parameters per cluster. This constrained CWM has no precedent in the CWM literature.

3. A **Koopman-theoretic variant** (local EDMD) where each cluster carries a continuous-time Koopman generator in a lifted polynomial space, replacing the affine local model with a richer dynamical surrogate.

4. **Statistically rigorous empirical characterization** (10 seeds, paired t-tests) on Lorenz and pendulum benchmarks, showing when piecewise methods help (non-polynomial systems, $\sim 10\times$ parameter savings, $p < 0.001$) and when they do not (polynomial systems where global EDMD is exact).

2 Background

2.1 Linearization and Taylor remainder

Given a smooth vector field $f : \mathbb{R}^d \to \mathbb{R}^d$, the Jacobian at x_0 is $\boldsymbol{J}(x_0) \in \mathbb{R}^{d\times d}$ with $[\boldsymbol{J}(x_0)]_{ij} = \partial f_i / \partial x_j(x_0)$. Taylor's theorem gives, for x near x_0:

$$f(x) = f(x_0) + \boldsymbol{J}(x_0)(x - x_0) + \tfrac{1}{2} H(\xi)(x - x_0, x - x_0), \tag{5}$$

for some ξ on the segment, where H is the Hessian tensor. The linearization error is $O(\|x - x_0\|^2)$. A single linearization is accurate in a ball of radius $r_0 \sim \varepsilon / \|H\|$ for target error ε.

2.2 Gaussian mixture models and EM

A GMM with N components has density $p(x) = \sum_k \pi_k \mathcal{N}(x; \mu_k, \boldsymbol{\Sigma}_k)$. Fitting by maximum likelihood is tractable via **Expectation-Maximization** (Dempster, Laird, Rubin 1977). Introducing latent assignments z_i:

E-step: compute soft responsibilities $r_{ik} = p(z_i = k \mid x_i)$.

M-step: update $(\pi_k, \mu_k, \boldsymbol{\Sigma}_k)$ to maximize expected complete-data log-likelihood.

EM is coordinate ascent on the Evidence Lower Bound $\mathrm{ELBO}(q, \theta) = \mathbb{E}_q[\log p(X, Z \mid \theta)] + H[q]$. Both E- and M-steps are non-decreasing in ELBO.

2.3 Koopman theory and EDMD

The Koopman operator $\mathcal{K}_t$ associated with a flow φ_t acts on observables $g : \mathbb{R}^d \to \mathbb{R}$ by $(\mathcal{K}_t g)(x) = g(\varphi_t(x))$. Though the flow is generally nonlinear, $\mathcal{K}_t$ is *linear* on observables; its infinitesimal generator $\mathcal{L}$ satisfies $(\mathcal{L}g)(x) = \nabla g(x) \cdot f(x)$. Extended Dynamic Mode Decomposition (EDMD; Williams, Kevrekidis, Rowley 2015) approximates $\mathcal{L}$ in a finite-dimensional subspace spanned by observables $\boldsymbol{\Phi}(x) = (\phi_1(x), \ldots, \phi_M(x))$ via weighted least squares: fit $\boldsymbol{M} \in \mathbb{R}^{M\times M}$ so that $(\nabla \boldsymbol{\Phi}(x_i)) \cdot f(x_i) \approx \boldsymbol{M} \cdot \boldsymbol{\Phi}(x_i)$. When the dictionary is Koopman-invariant, $\boldsymbol{M}$ is exact. For Lorenz (RHS of degree 2), a degree-2 monomial dictionary is Koopman-invariant and EDMD recovers the dynamics essentially exactly.

2.4 Normal-Inverse-Wishart conjugacy

For a Gaussian component with unknown $(c_k, \boldsymbol{\Sigma}_k)$, the conjugate prior is the NIW: $\boldsymbol{\Sigma}_k \sim \mathcal{IW}(\Psi_0, \nu_0)$, $c_k \mid \boldsymbol{\Sigma}_k \sim \mathcal{N}(\mu_0, \boldsymbol{\Sigma}_k / \kappa_0)$. The marginal likelihood is closed-form, enabling Bayesian selection of the cluster count N.

3 The Residual-Aware Framework

3.1 The generative model

$$\pi \sim \text{Dirichlet}(\alpha_0 \mathbf{1}_N), \tag{6}$$

$$\mathbf{\Sigma}_k \sim \mathcal{IW}(\Psi_0, \nu_0), \;\; c_k \mid \mathbf{\Sigma}_k \sim \mathcal{N}(\mu_0, \mathbf{\Sigma}_k/\kappa_0), \tag{7}$$

$$z_i \sim \text{Cat}(\pi), \;\; x_i \mid z_i = k \sim \mathcal{N}(c_k, \mathbf{\Sigma}_k), \tag{8}$$

$$f_{\text{obs}}(x_i) \mid x_i, z_i = k \sim \mathcal{N}(L_k(x_i), \sigma^2 I_d). \tag{9}$$

The last line embeds the residual factor. The observed $f_{\text{obs}}(x_i)$ is the true vector-field value at x_i (assumed known or numerically integrable).

Remark 1 (On input-dependent means)**.** The factor $\mathcal{N}(f_{\text{obs}}(x_i); L_k(x_i), \sigma^2 I)$ is a *conditional* Gaussian: given the input x_i and the assignment $z_i = k$, the output $f_{\text{obs}}(x_i)$ is a Gaussian random variable whose mean depends on x_i through $L_k(x_i)$. This is the same structure as linear regression ($y \mid x \sim \mathcal{N}(w^\top x, \sigma^2)$) or mixtures of experts. Equivalently, the *residual* $\varepsilon_k(x_i) = f_{\text{obs}}(x_i) - L_k(x_i)$ has a fixed distribution $\mathcal{N}(0, \sigma^2 I)$ that does not depend on x_i; the input-dependence is absorbed into the transformation from f_{obs} to ε. Only the data-dependent squared-error term $\|\varepsilon_k(x_i)\|^2$ enters the likelihood; the normalizer $-\frac{d}{2}\log(2\pi\sigma^2)$ is input-independent.

3.2 The joint likelihood

$$p(x_i, f(x_i)) = \sum_{k=1}^{N} \pi_k \cdot \mathcal{N}(x_i; c_k, \mathbf{\Sigma}_k) \cdot \mathcal{N}(f(x_i); L_k(x_i), \sigma^2 I). \tag{10}$$

3.3 The evidence lower bound

With variational posterior $r_{ik} = q(z_i = k)$:

$$\begin{aligned} \text{ELBO}(q, \theta) = &\sum_{i,k} r_{ik} \left[\log \pi_k + \log \mathcal{N}(x_i; c_k, \mathbf{\Sigma}_k) + \log \mathcal{N}(\varepsilon_k(x_i); 0, \sigma^2 I)\right] \\ &- \sum_{i,k} r_{ik} \log r_{ik} + \log p(\pi \mid \alpha_0) + \sum_k \log p(c_k, \mathbf{\Sigma}_k \mid \text{NIW}). \end{aligned} \tag{11}$$

The ELBO is monotone non-decreasing across E-M iterations provided each M-step is a true coordinate-ascent maximization. We discuss how each variant satisfies or approximates this condition in Section 4.

3.4 Contrast with standard GMM

In the limit $\sigma^2 \to \infty$, the residual term $-\|\varepsilon_k(x_i)\|^2/(2\sigma^2) \to 0$ and the framework reduces exactly to standard GMM. In the limit $\sigma^2 \to 0$, the residual dominates and the model becomes a piecewise regression. We calibrate σ^2 automatically from within-cluster residuals during initialization:

$$\sigma^2 \leftarrow \text{median}_{i,k}\big(\|\varepsilon_k(x_i)\|^2\big)/d. \tag{12}$$

4 Local-Model Variants

4.1 Variant A: Taylor-analytic

Parameterization: $L_k(x) = f(c_k) + \boldsymbol{J}(c_k)(x - c_k)$, where $\boldsymbol{J}$ is evaluated *analytically* at c_k. Zero free local-model parameters.

Intuition: Classical pointwise Taylor. Excellent local accuracy (zero error at $x = c_k$, $O(\|x - c_k\|^2)$ away), strongly regularized, no fitting noise. The downside: when the cluster is large, points at the boundary are poorly modeled because Taylor's theorem gives only local accuracy.

ELBO: The center update for c_k should include the residual gradient, but that gradient is tangled (when c_k moves, both $f(c_k)$ and $\boldsymbol{J}(c_k)$ change). We use a GEM-style update: proximity-only gradient for c_k, then re-evaluate $f(c_k), \boldsymbol{J}(c_k)$. This can cause small ELBO non-monotonicities at high N.

4.2 Variant B: Taylor-LS (Hybrid)

Parameterization: $L_k(x) = f_k + \boldsymbol{J}_k(x - c_k)$, where $(f_k, \boldsymbol{J}_k)$ are free parameters fit by weighted LS per cluster. Free parameters per cluster: $d + d^2$.

M-step for $(f_k, \boldsymbol{J}_k)$: Closed-form weighted LS. Let $\phi_i = [1, (x_i - c_k)^\top]^\top$, stack into $Z \in \mathbb{R}^{P\times(d+1)}$, $F \in \mathbb{R}^{P\times d}$, $W = \mathrm{diag}(r_{ik})$:

$$[f_k^\top \; \boldsymbol{J}_k^\top]^\top = (Z^\top W Z + \lambda I)^{-1} Z^\top W F. \tag{13}$$

M-step for c_k (full coordinate-ascent):

$$\left[\Lambda_0 + R_k \boldsymbol{\Sigma}_k^{-1} + (R_k/\sigma^2)\boldsymbol{J}_k^\top \boldsymbol{J}_k\right] c_k^{\text{new}} = \Lambda_0 \mu_0 + \boldsymbol{\Sigma}_k^{-1} S_x - \tfrac{1}{\sigma^2}\boldsymbol{J}_k^\top (S_f - R_k f_k - \boldsymbol{J}_k S_x), \tag{14}$$

where $R_k = \sum_i r_{ik}$, $S_x = \sum_i r_{ik} x_i$, $S_f = \sum_i r_{ik} f(x_i)$. Under this update the ELBO is monotone.

Crossover intuition: When clusters are small, $\boldsymbol{J}(c_k) \approx \boldsymbol{J}_k^{\text{LS}}$, and analytic has zero noise while LS has fitting noise—analytic wins. When clusters are large, cluster-averaged $\boldsymbol{J}_k^{\text{LS}}$ differs substantially from pointwise $\boldsymbol{J}(c_k)$, and averaging beats pointwise because the Taylor remainder dominates—LS wins.

4.3 Variant C: Local EDMD

Parameterization: $L_k(x) = [\boldsymbol{M}_k \cdot \boldsymbol{\Phi}(x - c_k)]_{1:d}$, where $\boldsymbol{\Phi}\colon \mathbb{R}^d \to \mathbb{R}^M$ is a polynomial lift of degree p, and $\boldsymbol{M}_k \in \mathbb{R}^{M\times M}$ is a local continuous Koopman generator. Parameters per cluster: $M^2 = \binom{d+p}{p}^2$. For $d = 3, p = 2$: $M = 10$, so $\boldsymbol{M}_k$ has 100 parameters.

Intuition: Each cluster carries its own locally-valid finite-dimensional Koopman subspace. For a small region of phase space, even a modest polynomial lift can form an approximately-invariant subspace, because nonlinearities of f are well-approximated by low-degree polynomials over narrow supports.

M-step for $\boldsymbol{M}_k$: Weighted continuous EDMD fit:

$$\boldsymbol{M}_k = \Big(\sum_i r_{ik}[\nabla\boldsymbol{\Phi}_i \cdot f_i]\boldsymbol{\Phi}_i^\top\Big) \cdot \Big(\sum_i r_{ik}\boldsymbol{\Phi}_i\boldsymbol{\Phi}_i^\top + \lambda I\Big)^{-1}. \tag{15}$$

The predicted vector field is $\hat{f}(x) = [\boldsymbol{M}_k \boldsymbol{\Phi}(x - c_k)]_{1:d}$ (the entries corresponding to $\dot{u}$).

Why this helps on non-polynomial systems: Global EDMD at degree p can only approximate $\sin(\theta)$ by a fixed degree-p Taylor polynomial, which degrades as $|\theta|$ grows. Local EDMD at degree p around θ_k does Taylor around θ_k, which is locally accurate per cluster.

4.4 Summary

Variant	Params/cluster	Narrative	Monotone ELBO	Works without $\boldsymbol{J}$?
Taylor-analytic	0	Physics-grounded Taylor	Approximate (GEM)	No
Taylor-LS	$d + d^2$	Data-driven affine fit	Yes (pure CA-EM)	Yes
Local EDMD	$\binom{d+p}{p}^2$	Lifted Koopman per cluster	Yes (fixed c_k)	Yes

Table 1: Local-model variants.

5 The EM Algorithm

5.1 Initialization

GMM warm start on $\{x_i\}$ alone. Hard labels initialize: (i) Taylor-analytic uses $f(c_k), \boldsymbol{J}(c_k)$ analytically; (ii) Taylor-LS fits $(f_k, \boldsymbol{J}_k)$ by weighted LS on hard labels; (iii) Local EDMD fits $\boldsymbol{M}_k$ by weighted continuous EDMD. σ^2 is then calibrated from residuals. We use 2–3 random seeds and select the best run by final ELBO.

5.2 The E-step (identical for all variants)

$$\log \tilde{r}_{ik} = \log \pi_k + \log \mathcal{N}(x_i; c_k, \boldsymbol{\Sigma}_k) + \log \mathcal{N}(\varepsilon_k(x_i); 0, \sigma^2 I), \tag{16}$$

$$\log r_{ik} = \log \tilde{r}_{ik} - \operatorname{logsumexp}_k \log \tilde{r}_{ik}. \tag{17}$$

5.3 M-step variants

See Algorithms 3 and 4. The Taylor-analytic M-step is identical in structure to Algorithm 3 but uses $f_k \leftarrow f(c_k^{\text{new}}), \boldsymbol{J}_k \leftarrow \boldsymbol{J}(c_k^{\text{new}})$ analytically and omits the residual terms in the c_k update (14).

5.4 Dead-cluster pruning

With $\alpha_0 < 1$ the Dirichlet prior drives $\pi_k \to 0$ for unused components. When $R_k < 1$ we permanently remove the cluster. The user starts with N larger than needed; pruning finds the right count.

5.5 Model selection

For the proximity factor alone, the NIW marginal likelihood is exact (closed-form):

$$\begin{aligned}\log p(X_k) = \tfrac{d}{2} \log(\kappa_0/\kappa_n) + \tfrac{\nu_0}{2} \log|\Psi_0| - \tfrac{\nu_n}{2} \log|\Psi_n| \\ + \log \Gamma_d(\tfrac{\nu_n}{2}) - \log \Gamma_d(\tfrac{\nu_0}{2}) - \tfrac{R_k d}{2} \log \pi,\end{aligned} \tag{18}$$

with $\kappa_n = \kappa_0 + R_k$, $\nu_n = \nu_0 + R_k$. However, the residual factor $p(F_k \mid X_k, c_k, \sigma^2)$ depends on c_k nonlinearly (through $f(c_k)$ and $\boldsymbol{J}(c_k)$), breaking NIW conjugacy. We use a **plug-in approximation**: evaluate the residual at the converged $\hat{c}_k$ rather than integrating it out:

$$\log \widetilde{p}(X_k, F_k) \approx \log p(X_k) - \tfrac{1}{2\sigma^2} \sum_i \|\varepsilon_k(x_i)\|^2 \Big|_{c_k = \hat{c}_k}. \tag{19}$$

This is adequate when the posterior on c_k is concentrated (large R_k) and reduces to the exact NIW marginal when $\sigma^2 \to \infty$.

6 Algorithms

Algorithm 1: *Unified EM loop*

Input: $X \in \mathbb{R}^{P\times d}, F \in \mathbb{R}^{P\times d}$; initial N; hyperparameters; MSTEP
Output: state, responsibilities r, history

state $\leftarrow$ INITIALIZE(X, F, N, hp)
history $\leftarrow [\,]$
for $t = 1, \ldots, n_{\mathrm{iter}}$ **do**
  $r \leftarrow$ ESTEP$(X, F, \mathrm{state}, \mathrm{hp})$
  state$, r \leftarrow$ PRUNEDEAD$(\mathrm{state}, r, X, F, \mathrm{hp})$
  if state.$N = 0$ **then break**
  history.append(ELBO$(X, F, r, \mathrm{state}, \mathrm{hp})$)
  if $|\mathrm{history}[-1] - \mathrm{history}[-2]| < \mathrm{tol}$ **then break**
  state $\leftarrow$ MSTEP$(X, F, r, \mathrm{state}, \mathrm{hp})$
end for
return state$, r,$ history

Algorithm 2: *E-step (common to all variants)*

Input: $X, F, \mathrm{state}, \mathrm{hp}$

$\log \pi \leftarrow \log \mathrm{state}.\pi$
$\log$-prox $\leftarrow$ mvn_logpdf$(X, \mathrm{state.centers}, \mathrm{state.covariances})$ $(P \times N)$
$\log$-res $\leftarrow$ residual_logpdf$(X, F, \mathrm{state}.L, \sigma^2)$ $(P \times N)$
$\log \tilde{r} \leftarrow \log \pi + \log\text{-prox} + \log\text{-res}$
$\log r \leftarrow \log \tilde{r} - \mathrm{logsumexp}(\log \tilde{r}, \mathrm{axis} = 1)$
return $\exp(\log r)$

Algorithm 3: *M-step (Taylor-LS variant)*

Input: $X, F, r, \mathrm{state}, \mathrm{hp}$

$R \leftarrow \mathrm{sum}(r, 0)$; $\boldsymbol{\Sigma}^{-1} \leftarrow \mathrm{inv}(\mathrm{state.covariances})$
for $k = 1, \ldots, N$ **do**
  $r_k \leftarrow r[:, k]$; $R_k \leftarrow R[k]$
  Step 1: weighted LS for $f_k, \boldsymbol{J}_k$
  $\delta \leftarrow X - c_k$; $Z \leftarrow [\mathbf{1}, \delta] \in \mathbb{R}^{P\times(d+1)}$
  $\beta \leftarrow \mathrm{solve}(Z^\top \mathrm{diag}(r_k) Z + \lambda I,\ Z^\top \mathrm{diag}(r_k) F)$
  $f_k \leftarrow \beta[0]$; $\boldsymbol{J}_k \leftarrow \beta[1{:}]^\top$
  Step 2: c_k *update (full gradient)*
  $S_x \leftarrow r_k^\top X$; $S_f \leftarrow r_k^\top F$
  $\mathrm{LHS} \leftarrow \Lambda_0 + R_k \boldsymbol{\Sigma}_k^{-1} + (R_k/\sigma^2)\boldsymbol{J}_k^\top \boldsymbol{J}_k$
  $\mathrm{RHS} \leftarrow \Lambda_0 \mu_0 + \boldsymbol{\Sigma}_k^{-1} S_x - (1/\sigma^2)\boldsymbol{J}_k^\top (S_f - R_k f_k - \boldsymbol{J}_k S_x)$
  $c_k^{\mathrm{new}} \leftarrow \mathrm{solve}(\mathrm{LHS}, \mathrm{RHS})$
  Step 3: $\boldsymbol{\Sigma}_k$ *posterior mode*
  $\mathrm{diff} \leftarrow X - c_k^{\mathrm{new}}$; $\mathrm{scatter} \leftarrow \mathrm{diff}^\top \mathrm{diag}(r_k) \mathrm{diff}$
  $\boldsymbol{\Sigma}_k^{\mathrm{new}} \leftarrow (\Psi_0 + \mathrm{scatter})/(\nu_0 + R_k + d + 1) + \lambda I$
end for
$\pi^{\mathrm{new}} \leftarrow (R + \alpha_0 - 1)/(P + N(\alpha_0 - 1))$; normalize
return state

Algorithm 4: *M-step (Local EDMD variant)*

Input: $X, F, r, \text{state}, \text{hp}, \text{exps}$ (monomial exponents)
$R \leftarrow \text{sum}(r, 0)$; $\mathbf{\Sigma}^{-1} \leftarrow \text{inv}(\text{state.covariances})$
for $k = 1, \ldots, N$ **do**
  $r_k \leftarrow r[:, k]$; $R_k \leftarrow R[k]$
  Step 1: c_k update (proximity-only)
  $\text{LHS} \leftarrow \Lambda_0 + R_k \mathbf{\Sigma}_k^{-1}$; $\text{RHS} \leftarrow \Lambda_0 \mu_0 + \mathbf{\Sigma}_k^{-1}(r_k^\top X)$
  $c_k^{\text{new}} \leftarrow \text{solve}(\text{LHS}, \text{RHS})$
  Step 2: weighted continuous EDMD for $\boldsymbol{M}_k$
  $U \leftarrow X - c_k^{\text{new}}$; $\mathbf{\Phi} \leftarrow \text{monomials}(U, \text{exps})$ $\quad (P \times M)$
  $\nabla \mathbf{\Phi} \leftarrow \text{monomials\_grad}(U, \text{exps})$ $\quad (P \times M \times d)$
  $\dot{\mathbf{\Phi}} \leftarrow \text{einsum}(\text{'pmd,pd} \rightarrow \text{pm'}, \nabla \mathbf{\Phi}, F)$
  $G \leftarrow \mathbf{\Phi}^\top \text{diag}(r_k) \mathbf{\Phi}$; $A \leftarrow \dot{\mathbf{\Phi}}^\top \text{diag}(r_k) \mathbf{\Phi}$
  $\boldsymbol{M}_k \leftarrow \text{solve}(G + \lambda I, A^\top)^\top$
  Step 3: $\mathbf{\Sigma}_k$ posterior mode
  $\text{diff} \leftarrow X - c_k^{\text{new}}$; $\text{scatter} \leftarrow \text{diff}^\top \text{diag}(r_k) \text{diff}$
  $\mathbf{\Sigma}_k^{\text{new}} \leftarrow (\Psi_0 + \text{scatter})/(\nu_0 + R_k + d + 1) + \lambda I$
end for
$\pi^{\text{new}} \leftarrow (R + \alpha_0 - 1)/(P + N(\alpha_0 - 1))$; normalize
return state

7 Experiments

7.1 Systems and metrics

Lorenz ($d = 3$): $\dot{x} = \sigma(y - x), \dot{y} = x(\rho - z) - y, \dot{z} = xy - \beta z$, $(\sigma, \rho, \beta) = (10, 28, 8/3)$. Chaotic, RHS of degree 2. Data: 5000 points at $\Delta t = 0.01$ after warmup; train/test split 4000/1000.

Damped pendulum ($d = 2$): $\dot{\theta} = \dot{\theta}, \ddot{\theta} = -\sin\theta - \gamma\dot{\theta}$, $\gamma = 0.2$. Dissipative, non-polynomial. Data: 4000 training points uniformly sampled from $(\theta, \dot{\theta}) \in [-\pi, \pi] \times [-3, 3]$; 1000 held-out test points.

Metrics: (i) one-step vector-field error $\text{mean}_i \|\hat{f}(x_i) - f(x_i)\|$; (ii) rollout error $\|\hat{x}_t - x_t\|$ from attractor initial conditions (true trajectory from RK45, learned via switching Euler); (iii) parameter count.

7.2 Lorenz results

1. Global EDMD dominates absolutely. Across 10 seeds, degree-2 achieves $2.3 \pm 0.4\%$ one-step relative error; degree-3 achieves $0.03 \pm 0.00\%$—essentially exact, as expected since Lorenz is a degree-2 polynomial system.

2. Ours beats GMM at every N, with high statistical significance. Improvement ranges from $+34\%$ at $N = 5$ to $+50\%$ at $N = 50$ (paired t-test $p < 0.001$ at all N). GMM saturates at $\sim$12% relative error regardless of N (it prunes to $\sim$20 active clusters), while our method continues to improve: $19.8 \pm 1.9\%$ at $N = 5$ down to $6.2 \pm 0.3\%$ at $N = 50$.

3. Local EDMD is constant in N. At any $N \geq 2$ and any degree ≥ 2, local EDMD produces identical $\sim$5.5% one-step error. Since the global Koopman is exact in a 10-dim lift, every local cluster learns the same polynomial. The gap between local EDMD (5.5%) and global EDMD (2.3%) is a data-starvation effect: each local operator is fit from $\sim P/N$ responsibility-weighted points with ridge regularization. Increasing the training set size closes this gap proportionally.

4. Rollout stability: At $N \geq 12$ all piecewise methods produce bounded rollouts; below that, GMM diverges numerically ($> 10^{10}$) while our method stays bounded from $N \geq 5$.

Interpretation: Lorenz is the wrong system to demonstrate clustering benefit over global EDMD. But the comparison against GMM is decisive: the residual-aware clustering is a strict, statistically significant improvement over geometry-only clustering at every N tested.

7.3 Pendulum results

Pendulum exhibits the opposite: $\sin(\theta)$ has no finite polynomial representation, so global EDMD must use high-degree lifts.

Method	N	Params	One-step	Rollout @ 10s	n
Global EDMD, deg=2	1	36	0.391 ± 0.004	0.894 ± 0.009	10
Global EDMD, deg=4	1	225	0.057 ± 0.001	0.270 ± 0.007	10
Global EDMD, deg=6	1	784	0.004 ± 0.000	0.180 ± 0.000	10
Global EDMD, deg=8	1	2025	$\mathbf{0.000 \pm 0.000}$	0.174 ± 0.000	10
local-EDMD d2, N=2	2	**72**	0.015 ± 0.000	$\mathbf{0.166 \pm 0.005}$	10
local-EDMD d2, N=8	8	288	0.007 ± 0.001	0.171 ± 0.003	10
Taylor-ana., N=8	8	**48**	0.021 ± 0.003	$\mathbf{0.143 \pm 0.007}$	10
Taylor-ana., N=16	16	**96**	0.009 ± 0.001	$\mathbf{0.149 \pm 0.004}$	10
Taylor-LS, N=2	2	12	0.372 ± 0.058	1.265 ± 0.516	10
Taylor-LS, N=8	8	48	0.022 ± 0.002	0.179 ± 0.017	10

Table 2: Pendulum results (mean ± 95% CI across 10 seeds, 5 rollout ICs each).

1. Parameter efficiency. Local-EDMD deg-2 at $N = 2$ (72 params) significantly outperforms global EDMD deg-6 (784 params) on rollout: 0.166 ± 0.005 vs 0.180 ± 0.000 rad (paired t-test $p = 0.0003$)—an $\sim 11\times$ parameter-efficiency gain.

2. Taylor-analytic is the overall winner given analytic J. At $N = 8$ (48 params), Taylor-analytic achieves 0.143 ± 0.007 rad rollout, significantly beating global EDMD deg-8 at $21\times$ fewer parameters ($p < 0.0001$). Taylor-analytic dominates Taylor-LS at $N \geq 8$ ($p = 0.0003$).

3. Taylor-LS is unstable at coarse N. At $N = 2$, Taylor-LS exhibits high variance (1.265 ± 0.516 rad) across seeds, while Taylor-analytic is more stable (0.724 ± 0.076 rad). A single-seed comparison suggested Taylor-LS was superior at coarse N; the multi-seed analysis reveals this was not robust. At coarse partitions, LS-fit Jacobians from few data points suffer from fitting noise that overwhelms any averaging benefit over the Taylor expansion.

4. Global EDMD saturates at degree $\sim$6–8: rollout error plateaus at ~ 0.17 rad (Euler discretization floor).

7.4 Visual comparison

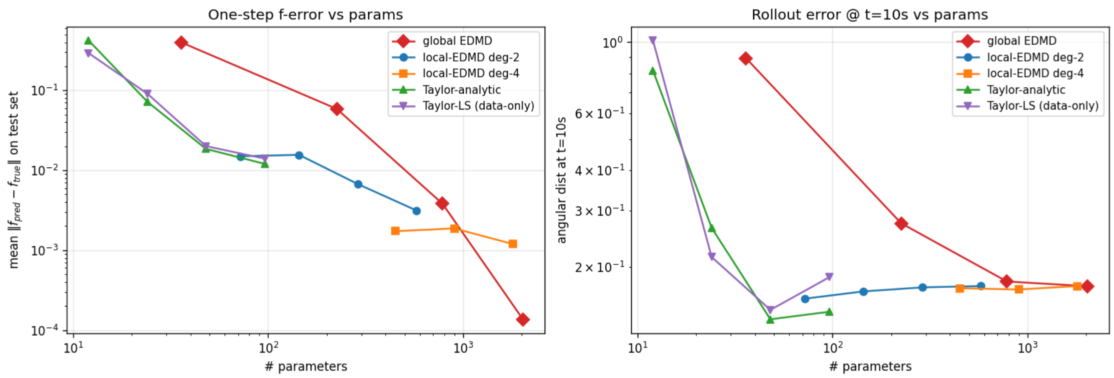

Figure 1: Pendulum: one-step error and rollout error vs. parameter count. Local-EDMD deg-2 and Taylor-analytic dominate the parameter-efficiency frontier.

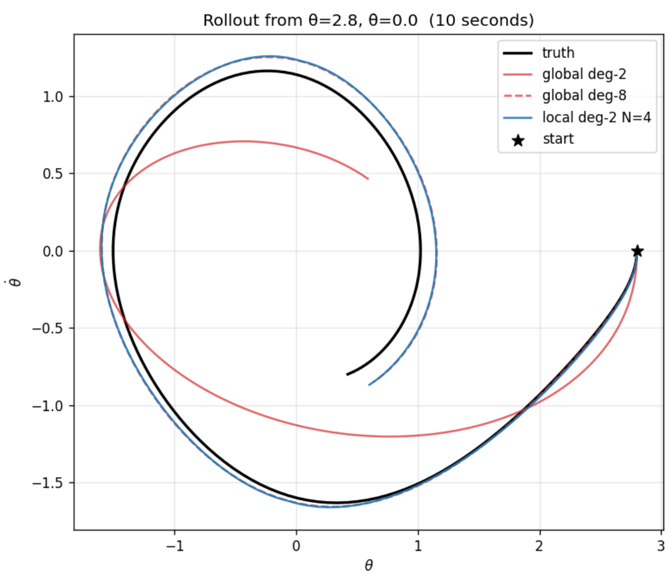

Figure 2: Rollout from $(\theta, \dot{\theta}) = (2.8, 0.0)$. Global EDMD deg-2 diverges; global EDMD deg-8 tracks truth; local EDMD deg-2 $N = 4$ is visually indistinguishable from global deg-8 at $\sim 14\times$ fewer parameters.

8 Discussion

8.1 When does each variant win?

System character	Analytic $f, \boldsymbol{J}$?	Best variant
Globally polynomial, low degree	—	Global EDMD (no partitioning)
Non-polynomial, physics known	Yes	Taylor-analytic
Non-polynomial, data-only	No	Taylor-LS or local-EDMD
Highly non-polynomial, rich local structure	No	Local-EDMD, degree ≥ 2

Table 3: Decision rule for variant selection.

8.2 Parameter efficiency

The framework's main practical selling point is parameter efficiency on non-polynomial systems. Global polynomial Koopman methods must use high degree to approximate $\sin, \exp, \text{sign}, \tanh$. The parameter count scales as $\binom{d+p}{p}^2$. Clustering into N regions with local degree-p lifts gives $N \cdot \binom{d+p}{p}^2$; per-cluster degree can be much smaller because each cluster's data span is narrower. We observed 10–20× savings on pendulum.

8.3 Limitations

Non-monotone ELBO at high N: Dead-cluster pruning is a discrete model-change step that drops ELBO by $O(\log N)$—an artifact, not a bug. The Taylor-analytic GEM structure can produce small ELBO drops.

Center update in local EDMD: Proximity-only; the residual gradient w.r.t. c_k involves $\nabla_{c_k} \boldsymbol{\Phi}(x - c_k)$, which is polynomial and algebraically heavy to close.

Cluster boundary handling: At test time we assign clusters by proximity alone (lacking $f(x)$ observations). This is a mild distribution shift.

Lyapunov horizon: For chaotic systems, rollout error grows exponentially regardless of model quality.

9 Related Work

Cluster-Weighted Models (Gershenfeld 1997 [5]; Gershenfeld, Schoner, Metois 1999 [6]; Ingrassia, Minotti, Vittadini 2012 [7]; Ingrassia, Minotti, Punzo 2014 [8]): our generative model is a CWM—the joint density $p(x, y) = \sum_k \pi_k \, p(x|k) \, p(y|x, k)$ with Gaussian components. The Linear Gaussian CWM [7] is mathematically equivalent to our Variant B (Taylor-LS) when $y = f(x)$. Punzo [9] extends CWM to polynomial local models, related to our local-EDMD variant. The CWM literature has focused on statistical clustering and classification (tourism, institutional evaluation); we extend it to dynamical systems with physics-constrained parameterizations and Koopman-theoretic local models.

Mixture of Experts (Jacobs, Jordan, Nowlan, Hinton 1991 [12]; Jordan & Jacobs 1994 [13]): MoE models the conditional $p(y|x) = \sum_k g_k(x) p(y|x, k)$ where gating $g_k(x)$ depends on x alone. CWM differs by also modeling $p(x|k)$, which regularizes clustering and enables Bayesian model selection.

Piecewise Affine System Identification (Ferrari-Trecate et al. 2003 [4]; Paoletti et al. 2007 [18]): two-stage approaches that cluster first (hard assignment, geometry-only) then fit local models. Our CWM-based approach performs clustering and regression simultaneously via EM with soft assignments, and the residual factor influences cluster formation—an advantage when geometric proximity does not imply linearizability.

Switching Linear Dynamical Systems (Fox et al. 2011 [10]; Ghahramani & Hinton 2000 [11]): regime-switching temporal models with HMM structure. We share the multiple-local-regimes premise but partition phase space rather than time.

EDMD and Koopman modeling (Williams et al. 2015 [19]; Klus et al. 2020 [15]; Brunton et al. 2022 [2]): fits Koopman operators globally. Our local-EDMD variant is cluster-wise EDMD with soft CWM-based assignments.

Our specific contributions beyond existing CWM: (i) a physics-constrained variant with analytic Jacobian (zero free local parameters); (ii) a Koopman-theoretic variant with lifted-space generators; (iii) first application of CWM to dynamical-systems phase-space partitioning; (iv) statistically rigorous comparison (10 seeds, paired t-tests) showing when each variant dominates.

10 Conclusion and Future Work

We have developed a residual-aware Bayesian clustering framework for piecewise local modeling of nonlinear vector fields. The framework generalizes standard GMM by introducing a joint likelihood over position and vector-field observation: regions are assigned based on both geometric proximity and local-model prediction accuracy. The local-model parameterization is modular, supporting physics-grounded Taylor (analytic Jacobian), data-driven affine regression (LS-fit), or Koopman lifting per cluster (local EDMD).

Three practically-important findings, validated across 10 random seeds with paired t-tests: (i) on globally-polynomial systems (Lorenz), partitioning does not improve on global EDMD, but residual-aware clustering is a strict improvement over GMM ($p < 0.001$ at every N tested, +34% to +50% improvement); (ii) on non-polynomial systems (pendulum), partitioning delivers $\sim 10\times$ parameter-efficiency gains over global EDMD ($p < 0.001$); (iii) when analytic Jacobians are available, the Taylor-analytic variant dominates at all N—the LS-fit variant, while conceptually appealing at coarse partitions, exhibits high variance across seeds and does not consistently outperform the analytic variant.

Future Work. (i) Stronger test systems: chaotic driven pendulum, coupled oscillators, neuronal models, stick-slip friction—settings with multiple distinct Koopman-invariant subspaces. (ii) Control-theoretic extensions: EDMDc per cluster. (iii) Automatic per-cluster lift degree selection. (iv) Rigorous ELBO handling for pruning events. (v) Non-isotropic residual covariance $\mathbf{\Sigma}_k^{\mathrm{res}}$.

Code and data. All implementations, validation scripts, and raw experiment data are available at https://github.com/agencyenterprise/fractal_residual_aware_clustering.

References

[1] A. Bemporad, G. Ferrari-Trecate, M. Muselli. Observability and controllability of piecewise affine and hybrid systems. *IEEE Trans. Aut. Control*, 46(12):1864–1876, 2001.

[2] S. L. Brunton, M. Budišić, E. Kaiser, J. N. Kutz. Modern Koopman theory for dynamical systems. *SIAM Review*, 64(2):229–340, 2022.

[3] A. P. Dempster, N. M. Laird, D. B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *J. Roy. Statist. Soc. B*, 39(1):1–38, 1977.

[4] G. Ferrari-Trecate, M. Muselli, D. Liberati, M. Morari. A clustering technique for the identification of piecewise affine systems. *Automatica*, 39(2):205–217, 2003.

[5] N. Gershenfeld. Nonlinear inference and cluster-weighted modeling. *Annals of the New York Academy of Sciences*, 808(1):18–24, 1997.

[6] N. Gershenfeld, B. Schoner, E. Metois. Cluster-weighted modelling for time-series analysis. *Nature*, 397(6717):329–332, 1999.

[7] S. Ingrassia, S. C. Minotti, G. Vittadini. Local statistical modeling via a cluster-weighted approach with elliptical distributions. *J. Classification*, 29(3):363–401, 2012.

[8] S. Ingrassia, S. C. Minotti, A. Punzo. Model-based clustering via linear cluster-weighted models. *Comput. Statist. Data Anal.*, 71:159–182, 2014.

[9] A. Punzo. Flexible mixture modeling with the polynomial Gaussian cluster-weighted model. *Statist. Modelling*, 14(3):257–291, 2014.

[10] E. B. Fox, E. B. Sudderth, M. I. Jordan, A. S. Willsky. Bayesian nonparametric inference of switching dynamic linear models. *IEEE Trans. Signal Proc.*, 59(4):1569–1585, 2011.

[11] Z. Ghahramani, G. E. Hinton. Variational learning for switching state-space models. *Neural Comput.*, 12(4):831–864, 2000.

[12] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, G. E. Hinton. Adaptive mixtures of local experts. *Neural Comput.*, 3(1):79–87, 1991.

[13] M. I. Jordan, R. A. Jacobs. Hierarchical mixtures of experts and the EM algorithm. *Neural Comput.*, 6(2):181–214, 1994.

[14] E. Kaiser, B. R. Noack, L. Cordier, A. Spohn, M. Segond, M. Abel, G. Daviller, J. Östh, S. Krajnović, R. K. Niven. Cluster-based reduced-order modelling of a mixing layer. *J. Fluid Mech.*, 754:365–414, 2014.

[15] S. Klus, F. Nüske, S. Peitz, J. H. Niemann, C. Clementi, C. Schütte. Data-driven approximation of the Koopman generator. *Physica D*, 406:132416, 2020.

[16] B. O. Koopman. Hamiltonian systems and transformation in Hilbert space. *PNAS*, 17(5):315–318, 1931.

[17] E. N. Lorenz. Deterministic nonperiodic flow. *J. Atmos. Sci.*, 20(2):130–141, 1963.

[18] S. Paoletti, A. L. Juloski, G. Ferrari-Trecate, R. Vidal. Identification of hybrid systems: A tutorial. *Eur. J. Control*, 13(2–3):242–260, 2007.

[19] M. O. Williams, I. G. Kevrekidis, C. W. Rowley. A data-driven approximation of the Koopman operator: Extending dynamic mode decomposition. *J. Nonlinear Sci.*, 25(6):1307–1346, 2015.